통계열역학 (2011년도 1학기)

과제 #1 정답 (**총 70 점**)

1. **(총 20 점, 각 5 점)**

(a) 등온가역 팽창이므로 $\delta w_{\rm rev} = -P_{ext}d\bar{V} = -Pd\bar{V}$ 이고,

$$\begin{aligned} w_{\rm rev} &= -\int_{\bar{V}_1}^{\bar{V}_2} P d\bar{V} \\ &= -\int_{\bar{V}_1}^{\bar{V}_2} \left(\frac{RT}{\bar{V}-b} - \frac{a}{\bar{V}^2}\right) d\bar{V} \\ &= -\left[RT\ln(\bar{V}-b) + \frac{a}{\bar{V}}\right]_{\bar{V}_1}^{\bar{V}_2} \\ &= -RT\big(\ln(\bar{V}_2-b) - \ln(\bar{V}_1-b)\big) - \frac{a}{\bar{V}_2} + \frac{a}{\bar{V}_1} \\ &= -RT\ln\left(\frac{\bar{V}_2-b}{\bar{V}_1-b}\right) - \frac{a}{\bar{V}_2} + \frac{a}{\bar{V}_1}. \end{aligned} \tag{1}$$

(b) 이 기체의 내부 에너지가 온도에만 의존하므로 등온가역 팽창에서 내부 에너지의 변화 $\Delta U = 0$ 이다. 그러므로, 흡수한 열 q 는

$$\Delta U = w_{\rm rev} + q_{\rm rev} = 0 \Rightarrow q_{\rm rev} = -w_{\rm rev} = RT\ln\left(\frac{\bar{V}_2-b}{\bar{V}_1-b}\right) + \frac{a}{\bar{V}_2} - \frac{a}{\bar{V}_1}.$$

(c) 가역 팽창이므로, 엔트로피의 변화량 ΔS 는

$$\Delta S = \frac{q_{\rm rev}}{T} = R\ln\left(\frac{\bar{V}_2-b}{\bar{V}_1-b}\right) + \frac{a}{T}\left(\frac{1}{\bar{V}_2} - \frac{1}{\bar{V}_1}\right).$$

(d) Helmholtz 자유에너지의 정의 $A = U - TS$ 로부터,

$$\begin{aligned} \Delta A &= \Delta U - S\Delta T - T\Delta S, \\ &= \Delta U - T\Delta S, \quad (\text{등온 팽창이므로, } \Delta T = 0) \end{aligned}$$

$\Delta U = 0$ 이고

$$\Delta S = R\ln\left(\frac{\bar{V}_2-b}{\bar{V}_1-b}\right) + \frac{a}{T}\left(\frac{1}{\bar{V}_2} - \frac{1}{\bar{V}_1}\right).$$

이므로,

$$\Delta A = 0 - TR\ln\left(\frac{\bar{V}_2-b}{\bar{V}_1-b}\right) + a\left(\frac{1}{\bar{V}_2} - \frac{1}{\bar{V}_1}\right) = -RT\ln\left(\frac{\bar{V}_2-b}{\bar{V}_1-b}\right) + a\left(\frac{1}{\bar{V}_2} - \frac{1}{\bar{V}_1}\right).$$

2. **(10 점)**

$$\begin{aligned}\frac{P}{k_BT} &= -\left(\frac{M}{V}\right)\ln\left(1-\frac{N}{M}\right), \\ &= -\left(\frac{M}{V}\right)\left(-\frac{N}{M}-\frac{1}{2}\left(\frac{N}{M}\right)^2-\frac{1}{3}\left(\frac{N}{M}\right)^3+\cdots\right) \\ &= \frac{N}{V}+\frac{V}{2M}\left(\frac{N}{V}\right)^2+\frac{V^2}{3M^2}\left(\frac{N}{V}\right)^3+\cdots. \end{aligned} \tag{2}$$

한편, 주어진 문제로부터 $P/(k_BT)$ 는 다음과 같이 주어지므로

$$\frac{P}{k_BT} = \frac{N}{V}+B_2\left(\frac{N}{V}\right)^2+B_3\left(\frac{N}{V}\right)^3+\cdots, \tag{3}$$

(2) 식과 (3) 식을 비교하면 B_2 와 B_3 의 값은 다음과 같다.

$$B_2 = \frac{V}{2M} \quad \text{and} \quad B_3 = \frac{V^2}{3M^2}.$$

3. **(총 20 점, 각 5 점)**

(a)

$$\begin{aligned} Q(N,V,T) &= \sum_{j=0}^{\infty} e^{-\beta E_j} \\ &= \sum_{j=0}^{\infty} e^{-\beta\epsilon j} \\ &= 1+e^{-\beta\epsilon}+e^{-2\beta\epsilon}+e^{-3\beta\epsilon}+\cdots \\ &= \frac{1}{1-e^{-\beta\epsilon}} \end{aligned} \tag{4}$$

(b)

$$\begin{aligned}
\langle E\rangle &= \sum_{j=0}^{\infty}\frac{E_j e^{-\beta E_j}}{Q} \\
&= \sum_{j=0}^{\infty}\frac{\epsilon j e^{-\beta\epsilon j}}{Q} \\
&= \frac{1}{Q}\Big(\epsilon e^{-\beta\epsilon}+2\epsilon e^{-2\beta\epsilon}+3\epsilon e^{-3\beta\epsilon}+\cdots\Big) \\
&= \frac{1}{Q}\epsilon e^{-\beta\epsilon}\Big(1+2e^{-\beta\epsilon}+3e^{-2\beta\epsilon}+\cdots\Big) \\
&= \frac{1}{1/(1-e^{-\beta\epsilon})}\frac{\epsilon e^{-\beta\epsilon}}{(1-e^{-\beta\epsilon})^2} \qquad ((4)\text{ 식과 힌트로부터}) \\
&= \frac{\epsilon e^{-\beta\epsilon}}{1-e^{-\beta\epsilon}} \qquad (5)
\end{aligned}$$

$\langle E\rangle = a\epsilon$ 0이므로 (5) 식으로부터

$$a\epsilon = \frac{\epsilon e^{-\beta\epsilon}}{1-e^{-\beta\epsilon}} \Rightarrow a = \frac{e^{-\beta\epsilon}}{1-e^{-\beta\epsilon}} \Rightarrow e^{-\beta\epsilon} = \frac{a}{1+a} \qquad (6)$$

(6) 식으로부터

$$\begin{aligned}
&-\beta\epsilon = \ln\Big(\frac{a}{1+a}\Big) \Rightarrow \beta = -\frac{1}{\epsilon}\ln\Big(\frac{a}{1+a}\Big) \\
&\therefore \beta = \frac{1}{k_B T} = \frac{1}{\epsilon}\ln\Big(1+\frac{1}{a}\Big) \qquad (7)
\end{aligned}$$

(c) (7) 식으로부터

$$\begin{aligned}
T &= \frac{\epsilon}{k_B}\frac{1}{\ln\big(1+1/a\big)} \\
&= \frac{hc\bar{\nu}}{k_B}\frac{1}{\ln\big(1+1/a\big)} \\
&= \frac{(6.62608\times10^{-34}\ \text{J s})(2.9979\times10^{8}\ \text{m s}^{-1})(50\ \text{cm}^{-1})}{(1.38065\times10^{-23}\ \text{J K}^{-1})}\frac{1}{\ln 2} \\
&= \frac{(6.62608\times10^{-34}\ \text{J s})(2.9979\times10^{8}\ \text{m s}^{-1})(50\times10^{2}\ \text{m}^{-1})}{(1.38065\times10^{-23}\ \text{J K}^{-1})}\frac{1}{\ln 2} \\
&= 103.8\ \text{K} \qquad (8)
\end{aligned}$$

(d) 엔트로피의 정의는 다음과 같이 주어진다.

$$S = -k_B\sum_{j=0}^{\infty} p_j \ln p_j \qquad (9)$$

$p_j = e^{-\beta E_j}/Q = e^{-\beta\epsilon j}/Q$ 이므로 이를 (9) 식에 대입하면

$$\begin{aligned} S &= -k_B \sum_{j=0}^{\infty} p_j \ln p_j \\ &= -k_B \sum_{j=0}^{\infty} \frac{e^{-\beta\epsilon j} \ln(e^{-\beta\epsilon j}/Q)}{Q} \\ &= -k_B \sum_{j=0}^{\infty} \frac{e^{-\beta\epsilon j}(-\beta\epsilon j - \ln Q)}{Q} \\ &= k_B \sum_{j=0}^{\infty} \frac{e^{-\beta\epsilon j}(\beta\epsilon j)}{Q} + k_B \sum_{j=0}^{\infty} \frac{e^{-\beta\epsilon j} \ln Q}{Q} \\ &= k_B \beta \langle E \rangle + k_B \ln Q \end{aligned} \tag{10}$$

$\langle E \rangle = a\epsilon$ 이고 (6) 식으로부터

$$a = \frac{e^{-\beta\epsilon}}{1 - e^{-\beta\epsilon}} \Rightarrow Q = \frac{1}{1 - e^{-\beta\epsilon}} = a e^{\beta\epsilon} \tag{11}$$

$\langle E \rangle = a\epsilon$ 와 (11) 식을 (10) 식에 대입하면

$$\begin{aligned} S &= k_B \Big(\beta a\epsilon + \ln(a e^{\beta\epsilon}) \Big) \\ &= k_B \Big(a\beta\epsilon + \ln a + \beta\epsilon \Big) \\ &= k_B \left(a \ln\left(1 + \frac{1}{a}\right) + \ln a + \ln\left(1 + \frac{1}{a}\right) \right) \\ &= k_B \Big(a \ln(1+a) - a \ln a + \ln a + \ln(1+a) - \ln a \Big) \\ &= k_B \Big((1+a)\ln(1+a) - a \ln a \Big) \end{aligned} \tag{12}$$

4. **(총 10 점, 각 5 점)**

(a) 평균에너지 $\langle E \rangle$ 는

$$\begin{aligned} \langle E \rangle &= k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V} \\ &= k_B T^2 \frac{\partial}{\partial T} \ln \left[\frac{1}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} (V - Nb)^N e^{aN^2/(V k_B T)} \right] \\ &= k_B T^2 \frac{\partial}{\partial T} \left\{ -\ln N! + \frac{3N}{2} \ln \left(\frac{2\pi m k_B}{h^2} \right) + \frac{3N}{2} \ln T + N \ln(V - Nb) + \frac{aN^2}{V k_B T} \right\} \\ &= \frac{3}{2} N k_B T - \frac{aN^2}{V} \end{aligned}$$

C_V 는

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T}\right)_{N,V} = \frac{3}{2} N k_B$$

(b)

$$\begin{aligned}
P &= -\left(\frac{\partial A}{\partial V}\right)_{N,T} \\
&= k_B T \left(\frac{\partial \ln Q}{\partial V}\right)_{N,T} \\
&= k_B T \frac{\partial}{\partial V} \left\{ \ln \left[\frac{1}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} (V - Nb)^N e^{aN^2/(V k_B T)} \right] \right\} \\
&= k_B T \frac{\partial}{\partial V} \left\{ -\ln N! + \frac{3N}{2} \ln \left(\frac{2\pi m k_B T}{h^2} \right) + N \ln(V - Nb) + \frac{aN^2}{V k_B T} \right\} \\
&= k_B T \left\{ \frac{N}{V - Nb} - \frac{aN^2}{V^2 k_B T} \right\} \\
&= \frac{N k_B T}{V - Nb} - \frac{aN^2}{V^2}
\end{aligned}$$

5. **(총 10 점, 각 5 점)**

(a)

$$\begin{aligned}
\Delta \bar{H}^\circ &= \bar{H}^\circ(1000 \text{ K}) - \bar{H}^\circ(298.15 \text{ K}) \\
&= \int_{298.15 \text{ K}}^{1000 \text{ K}} \bar{C}_P(T) dT, \\
&= \int_{298.15 \text{ K}}^{1000 \text{ K}} (\alpha + \beta T + \gamma T^2 + \delta T^3) dT, \\
&= \left[\alpha T + \frac{\beta}{2} T^2 + \frac{\gamma}{3} T^3 + \frac{\delta}{4} T^4 \right]_{298.15 \text{ K}}^{1000 \text{ K}} \\
&= \alpha\big((1000 \text{ K}) - (298.15 \text{ K})\big) + \frac{\beta}{2}\big((1000 \text{ K})^2 - (298.15 \text{ K})^2\big) \\
&\quad + \frac{\gamma}{3}\big((1000 \text{ K})^3 - (298.15 \text{ K})^3\big) + \frac{\delta}{4}\big((1000 \text{ K})^4 - (298.15 \text{ K})^4\big) \qquad (13)
\end{aligned}$$

i. O_2(g)

$$
\begin{aligned}
\Delta\bar{H}^{\circ}(\mathrm{O_2(g)}) &= (25.460\ \mathrm{J\ K^{-1}\ mol^{-1}})\big((1000\ \mathrm{K}) - (298.15\ \mathrm{K})\big) \\
&\quad + \frac{(1.519 \times 10^{-2}\ \mathrm{J\ K^{-2}\ mol^{-1}})}{2}\big((1000\ \mathrm{K})^2 - (298.15\ \mathrm{K})^2\big) \\
&\quad + \frac{(-0.715 \times 10^{-5}\ \mathrm{J\ K^{-3}\ mol^{-1}})}{3}\big((1000\ \mathrm{K})^3 - (298.15\ \mathrm{K})^3)\big) \\
&\quad + \frac{(1.311 \times 10^{-9}\ \mathrm{J\ K^{-4}\ mol^{-1}})}{4}\big((1000\ \mathrm{K})^4 - (298.15\ \mathrm{K})^4\big) \\
&= 2.28 \times 10^4\ \mathrm{J\ mol^{-1}}
\end{aligned}
$$

ii. $\mathrm{CO(g)}$

$$
\begin{aligned}
\Delta\bar{H}^{\circ}(\mathrm{CO(g)}) &= (28.142\ \mathrm{J\ K^{-1}\ mol^{-1}})\big((1000\ \mathrm{K}) - (298.15\ \mathrm{K})\big) \\
&\quad + \frac{(0.167 \times 10^{-2}\ \mathrm{J\ K^{-2}\ mol^{-1}})}{2}\big((1000\ \mathrm{K})^2 - (298.15\ \mathrm{K})^2\big) \\
&\quad + \frac{(0.537 \times 10^{-5}\ \mathrm{J\ K^{-3}\ mol^{-1}})}{3}\big((1000\ \mathrm{K})^3 - (298.15\ \mathrm{K})^3)\big) \\
&\quad + \frac{(-2.221 \times 10^{-9}\ \mathrm{J\ K^{-4}\ mol^{-1}})}{4}\big((1000\ \mathrm{K})^4 - (298.15\ \mathrm{K})^4\big) \\
&= 2.17 \times 10^4\ \mathrm{J\ mol^{-1}}
\end{aligned}
$$

iii. $\mathrm{CO_2(g)}$

$$
\begin{aligned}
\Delta\bar{H}^{\circ}(\mathrm{CO_2(g)}) &= (22.243\ \mathrm{J\ K^{-1}\ mol^{-1}})\big((1000\ \mathrm{K}) - (298.15\ \mathrm{K})\big) \\
&\quad + \frac{(5.977 \times 10^{-2}\ \mathrm{J\ K^{-2}\ mol^{-1}})}{2}\big((1000\ \mathrm{K})^2 - (298.15\ \mathrm{K})^2\big) \\
&\quad + \frac{(-3.499 \times 10^{-5}\ \mathrm{J\ K^{-3}\ mol^{-1}})}{3}\big((1000\ \mathrm{K})^3 - (298.15\ \mathrm{K})^3)\big) \\
&\quad + \frac{(7.464 \times 10^{-9}\ \mathrm{J\ K^{-5}\ mol^{-1}})}{4}\big((1000\ \mathrm{K})^4 - (298.15\ \mathrm{K})^4\big) \\
&= 3.33 \times 10^4\ \mathrm{J\ mol^{-1}}
\end{aligned}
$$

(b) 안 풀어도 됨.